

Salifort Motors: Employee Retention Analysis

Google Advanced Data Analytics Capstone | Course 6

OVERVIEW

We've got a dataset of nearly 15,000 employees, with 10 variables to work with. After cleaning it up, we're left with about 12,000 records and no missing values. **Goal:** Figure out which employees are likely to leave and what's driving that decision. **Approach:** Built logistic regression, decision tree, and random forest classifiers. In the second round, we dropped `satisfaction_level` and added an `overworked` feature (`>175 hrs/month`) to avoid data leakage.

PROJECT STATUS

Champion Model: Random Forest (Round 2)

The model we ended up with can predict employee departure with pretty high accuracy — 96.2% accuracy and an AUC of 93.8%. This means HR can identify employees who are at risk of quitting before it happens.

Model Comparison:

Logistic Regression: 80% precision | 83% recall | 82% accuracy
Decision Tree Rd.2: 87% precision | 90.4% recall | 93.8% AUC

★ **Random Forest Rd.2: 87% precision | 90.4% recall | 96.2% accuracy | 93.8% AUC**

Limitations: Dropped `satisfaction_level` (leakage risk). Self-reported data. FP (67) > FN (48) — safer for retention use.

KEY INSIGHTS

Top 4 Predictors (Random Forest):

- 1. `last_evaluation`
- 2. `number_project`
- 3. `tenure`
- 4. `overworked (>175 hrs/mo.)`

- Any employee with 7 projects is likely to leave
- 240–315 hrs/month is basically a burnout cluster
- Employees at the 4-year mark tend to have really low satisfaction
- Almost no employees who got promoted ended up leaving
- The problem isn't specific to one department — it's company-wide

It's worth noting that these results should inform company-wide policy changes, not individual employee scores.

NEXT STEPS

1. Cap projects at 5 per employee

Anyone with 7 projects honestly, is at risk of leaving. Reassign overloaded staff immediately.

2. Check in with employees at the 3-year mark

See how they're doing and what they need to develop before the 4-year satisfaction drop hits.

3. Fix the imbalance between hours worked and rewards

Either fairly compensate overtime or reduce the expectation. Make it transparent.

4. Promote more employees

It seems to make a big difference in whether they stay or go. Almost no promoted employees left.

5. Next modeling steps:

Remove `last_evaluation` to eliminate leakage risk, test XGBoost, explore K-means clustering.